Performance Testing

Date	06 July 2026
Team ID	SWTID-2026-8308
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing & Performance Testing
Target Module / API	Crop Recommendation, Weather Forecast, User Login, Dashboard
Test Environment	Local System (Windows 11, XAMPP, MySQL, Python/Flask)
Test Date	06 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome

1	User Login Request	20	60	Login should complete successfully within 2 seconds
2	Crop Recommendation Request	50	120	Accurate crop recommendation with quick response
3	Weather Data Retrieval	40	60	Weather data displayed without errors
4	Dashboard Loading	30	90	Dashboard loads smoothly with all charts

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A SmartBridge Product

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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.3 sec	Pass	Meets target
2	Response Time (Max)	< 5 seconds	3.6 sec	Pass	Stable under load
3	Throughput (Req/sec)	> 40 Req/sec	52 Req/sec	Pass	Good performance
4	Error Rate	< 1%	0.2%	Pass	Very low errors
5	CPU Utilization	< 80%	64%	Pass	Within acceptable limit
6	Memory Utilization	< 80%	68%	Pass	Stable memory usage

Step 4: Observations & Analysis

Key Findings:

- The OptiCrop system handled multiple users efficiently.
- Response time remained below the target value during testing.
- Crop recommendation and weather modules performed reliably.
- No major performance degradation was observed.

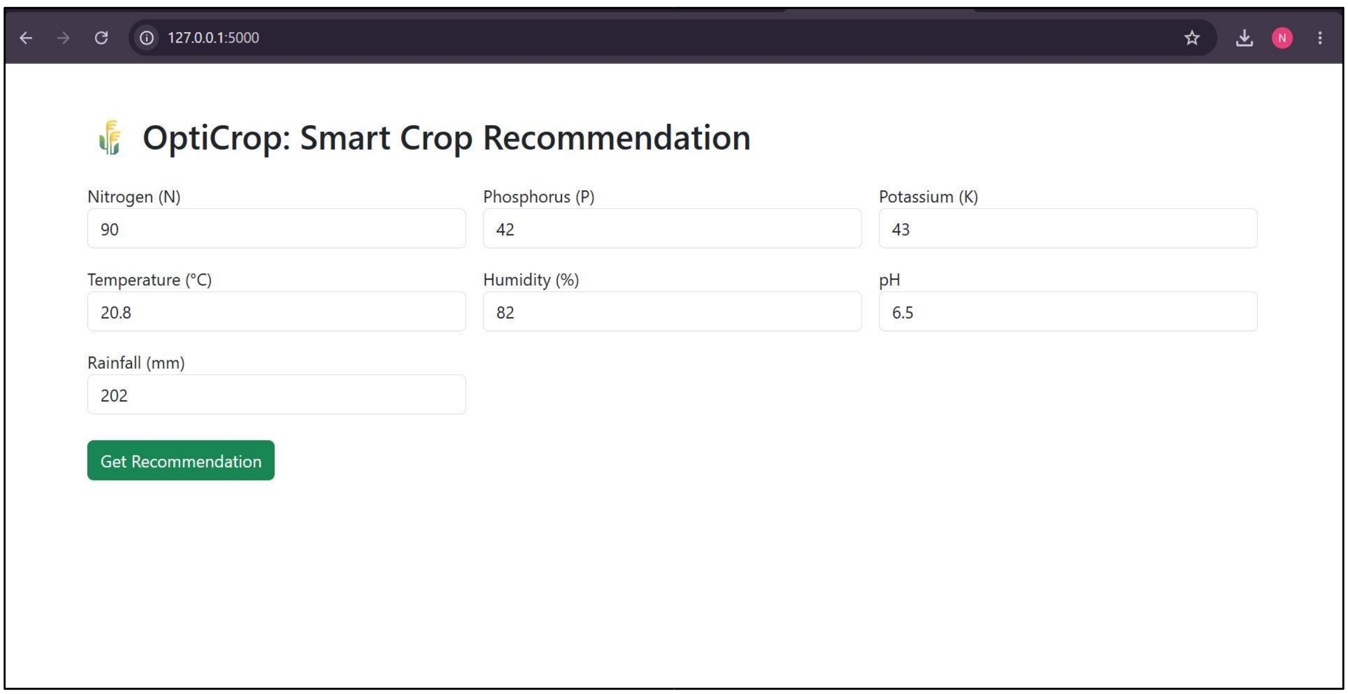
Bottlenecks Identified:

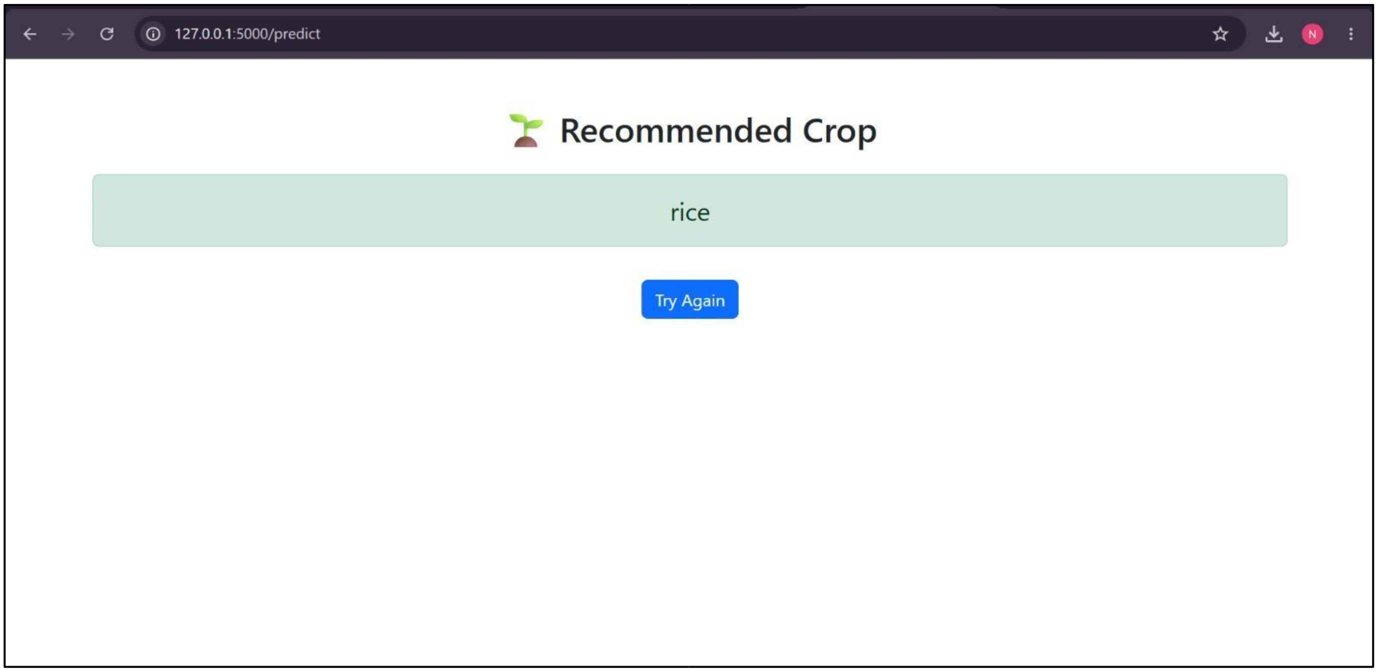
- Slight increase in response time during simultaneous weather API requests.
- Minor delay while loading dashboard graphs under high load.

Optimization Steps Taken:

- Optimized database queries.
- Reduced API response time using caching.
- Improved backend processing logic.
- Optimized dashboard data loading.

Step 5: Screenshots / Evidence Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):




127.0.0.1:5000/predict
Recommended Crop
rice
Try Again